

Supplementary Material: Full Proofs and Complete Baseline Table

D2T-RNA Project

1 Setup and notation

We use the registered finite model of Definition ?? in the main text: finite latent states $\Theta = \Theta_0 \sqcup \Theta_1$ (disjoint, so $\Theta_0 \cap \Theta_1 = \emptyset$), finite actions A , finite categorical alphabet $\mathcal{Y}$, passive marginal map M , action-level laws B_u , registered nuisance coupling $\mathcal{C}$, fixed non-adaptive horizon h , and abstention rule $\mathcal{R}$. The complete cross-class difference set is $D = \Theta_0 \times \Theta_1$, which is finite and contains no trivial diagonal pair (θ, θ) . For a panel $S \subseteq A$, the induced product law of θ is $P_S(\theta) = \bigotimes_{u \in S} B_u(\theta)$, authorized by $\mathcal{C}$, and $\gamma(S) = \min_{(\theta_0, \theta_1) \in D} \text{TV}(P_S(\theta_0), P_S(\theta_1))$ is the *observation-probability* separation (total-variation over $\mathcal{Y}^{|S|}$, not a state-space metric). Because D is finite and the laws are finite categorical laws, the minimum is attained, so every collision witness is non-trivial and attained.

We assume (Assumptions 1–2): the model is pre-registered, and the action-level likelihoods are complete and positive.

2 Proof of Theorem 1 (T2b: exact collision-or-separation)

Theorem 1 (T2b, restated). *Under the setup of Section 1, for any panel $S \subseteq A$:*

1. $\gamma(S) = 0$ iff there exists $(\theta_0, \theta_1) \in D$ with $P_S(\theta_0) = P_S(\theta_1)$;
2. $\gamma(S) > 0$ iff $P_S(\theta_0) \neq P_S(\theta_1)$ for all $(\theta_0, \theta_1) \in D$.

When $\gamma(S) > 0$, the realizing pair yields a rational primal/dual certificate and an independent-checker verification.

Proof. Existence of the minimum. The map $F : (\theta_0, \theta_1) \mapsto \text{TV}(P_S(\theta_0), P_S(\theta_1))$ takes values in the finite set $\{0, \frac{k}{L} : 0 \leq k \leq L\}$ for $L = |\mathcal{Y}|^{|S|}$, because each $B_u(\theta)$ is a probability vector over the finite alphabet $\mathcal{Y}$ and the product law is a finite categorical law. Hence F attains a minimum over the finite set D .

(i) *Forward direction.* Suppose $\gamma(S) = 0$. Then some pair (θ_0, θ_1) satisfies $\text{TV}(P_S(\theta_0), P_S(\theta_1)) = 0$. Total variation vanishes iff the two probability laws are identical, so $P_S(\theta_0) = P_S(\theta_1)$, an exact collision witness. This proves existence. *Reverse direction.* If $P_S(\theta_0) = P_S(\theta_1)$ for some pair, then $\text{TV}(P_S(\theta_0), P_S(\theta_1)) = 0$, so the minimum $\gamma(S)$ is 0.

(ii) *Forward direction.* Suppose $\gamma(S) > 0$. Since the minimum over D is attained and is positive, every pair has strictly positive TV, so $P_S(\theta_0) \neq P_S(\theta_1)$ for all pairs. *Reverse direction.* If all pairs have distinct laws, each has positive TV; the finite minimum is then positive, so $\gamma(S) > 0$.

This completes the iff directions. Finiteness of D and of the observation laws means every object in the proof is rational; the realizing pair is an explicit witness. The independent checker re-derives the collision (`verify_collision`) or separation (`verify_separation`) from first principles, and the LP strong-duality plus primal-feasibility check (`lp.py`) supplies the rational primal/dual pair. This concludes the proof. $\square$

3 Proof of Theorem 2 (T2c: finite-sample decision bound)

Theorem 2 (T2c, restated). *For a finite registered pair catalog under fixed-horizon deterministic non-adaptive allocation with complete action-level likelihoods and registered product laws:*

1. (*Upper bound*) $\mathbb{P}(\text{wrong}) \leq \frac{1}{2} \exp(-I_{\text{tot}})$ and $\mathbb{P}(\text{correct}) \geq 1 - \frac{1}{2} \exp(-I_{\text{tot}}) - \rho$;
2. (*Lower bound / structural no-go*) a correct-declaration floor κ requires $I_{\text{tot}} \geq -\frac{1}{2} \log(1 - (2\kappa - 1)^2)$.

Proof. For a fixed pair (θ_0, θ_1) and action u , write $p_0(y) = B_u(\theta_0)(y)$ and $p_1(y) = B_u(\theta_1)(y)$, and let $\rho_u = \sum_y \sqrt{p_0(y)p_1(y)}$ be the Bhattacharyya coefficient. The Hellinger information is $I_u = -\log \rho_u$.

(i) *Upper bound.* Under the registered product law over n_u i.i.d. observations of action u , the likelihood ratio between θ_0 and θ_1 has the standard Chernoff/Bhattacharyya concentration. For a decision rule that is the likelihood-ratio test (or any rule with minimax error at least as good as the trivial guess), the probability of declaring the wrong class satisfies

$$\mathbb{P}(\text{wrong}) \leq \frac{1}{2} \rho_u^{n_u} = \frac{1}{2} e^{-n_u I_u}.$$

Independence across actions in the product law gives $\mathbb{P}(\text{wrong}) \leq \frac{1}{2} \prod_u e^{-n_u I_u} = \frac{1}{2} e^{-I_{\text{tot}}}$ with $I_{\text{tot}} = \sum_u n_u I_u$. Since the decision-or-abstain rule either declares correctly, declares wrongly, or abstains, and $\mathbb{P}(\text{abstain}) = \rho$, we obtain $\mathbb{P}(\text{correct}) \geq 1 - \frac{1}{2} e^{-I_{\text{tot}}} - \rho$.

(ii) *Lower bound / structural no-go.* The decision error of the optimal two-class test between two product laws μ_0, μ_1 is at least $\frac{1}{2}(1 - \text{TV}(\mu_0, \mu_1))$, and the total-variation certainty can be bounded below only if the per-action coefficients are separated. Specifically, the requirement $\mathbb{P}(\text{correct}) \geq \kappa$ forces the Bhattacharyya product to be small enough that

$$\rho_{\text{tot}} \leq 1 - (2\kappa - 1)^2,$$

where $\rho_{\text{tot}} = \prod_u \rho_u^{n_u} = e^{-I_{\text{tot}}}$. Taking the log of both sides and rearranging yields $I_{\text{tot}} \geq -\frac{1}{2} \log(1 - (2\kappa - 1)^2)$. If the certified value of I_{tot} (from certified interval arithmetic) is strictly below this threshold, the floor is unachievable by any allocation within the budget, which constitutes the structural no-go. This completes the proof. $\square$

Tightness microcase. For alphabet size 2, $P_0 = (1/4, 3/4)$, $P_1 = (1, 0)$: $\rho_u = \sqrt{1/4 \cdot 1} + \sqrt{3/4 \cdot 0} = 1/2$, $I_u = \ln 2$. The exact equal-prior Bayes average error for this pair (computed by exhaustive enumeration) is $\frac{1}{2}(1/4)^n$, and the certified upper bound $\frac{1}{2} e^{-n I_u} = \frac{1}{2} (1/2)^n$ brackets the Bhattacharyya ideal. Hence the certified bound dominates the exact equal-prior oracle error for every $n \geq 1$.

4 Proof of Theorem 3 (T2d: costed design and no-go)

Theorem 3 (T2d, restated). *Consider the costed design problem (Definition in main text) over a finite registered pair catalog.*

1. (*Achievable upper*) an integer n^* with $\sum_u n_u^* I_{uw}^{\text{lo}} \geq \tau_w$ gives achievable cost $c^T n^*$;
2. (*No-go lower*) a dual-feasible y with $\tau^T y > \text{budget}$ rules out all feasible designs;
3. (*Integrality gap*) $\text{gap} = (\text{upper cost} - \text{lower bound}) / \text{lower bound}$.

Proof. (i) By construction, any integer vector n^* satisfying the coverage constraints is feasible for the primal. Its cost $c^T n^*$ is therefore an achievable upper bound. Feasibility is re-derivable from first principles (`check_integer_design_feasible`).

(ii) Consider the LP relaxation with $n_u \geq 0$ real. Its dual is $\max_{y \geq 0} \tau^T y$ subject to I_{uw}^{up} -based constraints; using the certified *upper* information bounds $I_{uw}^{\text{up}} \geq I_{uw}$ keeps each dual constraint valid for the true problem (for the true information $I_{uw} \leq I_{uw}^{\text{up}}$, the constraint $\sum_w y_w I_{uw}^{\text{up}} \leq c_u$ implies $\sum_w y_w I_{uw} \leq c_u$), so the relaxation is a valid underestimate. By weak duality, for every feasible integer n and every dual-feasible y ,

$$\sum_u c_u n_u \geq \sum_w \tau_w y_w.$$

If $\tau^T y > \text{budget}$, then every feasible design has cost strictly above the budget, so no feasible design within budget exists: a design-class no-go certificate. (Necessity/no-go therefore uses the *upper* information bound.)

(iii) The integrality gap quantifies the gap between the achievable integer upper cost and the dual lower bound; with certified interval information it is computed conservatively – using the *upper* information bound for the no-go/lower direction and the *lower* information bound for the constructive achievability/upper direction (an integer n with $\sum_u n_u I_{uw}^{\text{lo}} \geq \tau_w$ is guaranteed feasible since $I_{uw} \geq I_{uw}^{\text{lo}}$). This completes the proof. $\square$

Exact costed microcases. Single-pair integral: $\min n$, $n \geq 3$, cost 2: IP=LP=6, gap 0. Fractional cover 3×3 with $A = [[1, 0, 1], [1, 1, 0], [0, 1, 1]]$, $\tau = (1, 1, 1)$, $c = (1, 1, 1)$: LP=3/2, IP=2, gap=1/3. Fractional requirement $\min n$ with $n \geq 3/2$, cost 1: LP=3/2, IP=2, gap=1/3. No-go demonstration: fractional cover under budget $7/5 < 3/2$ is NO_GO.

5 Registered real-data case: add riboswitch

5.1 Provenance and registration

The add adenine-riboswitch aptamer is 71 nt (residues 13–83 of PDB 1Y26). We register two latent states: *OFF* (apo ensemble, represented by the ViennaRNA MFE dot-bracket) and *ON* (adenine-bound, base pairs read from PDB 1Y26). $\theta_0 = e_{\text{OFF}} = (1, 0)$, $\theta_1 = e_{\text{ON}} = (0, 1)$, passive marginal $M = (1, 1)$ (so $M\theta_0 = M\theta_1 = 1$, a genuine passive collision fiber), and one binary protected/reactivity action per position (71 actions). The registered measurement-noise coupling $\mathcal{C}$ is a symmetric bit-flip channel with error $\varepsilon \in (0, 1/2)$; $\varepsilon = 1/10$ is the default and is swept in the runner. Because $\varepsilon > 0$, the Bhattacharyya coefficient is positive and the per-position Hellinger information is finite. Of the 71 positions, 21 are separating (bare pairing differs between OFF and ON).

5.2 T2b certificate

For the passive panel $S = \emptyset$, $\gamma = 0$ (collision witness). For any panel containing a separating position, $\gamma(S) = |1 - 2\varepsilon|$ in observation-probability total-variation norm: $\gamma = 9/10$ at $\varepsilon = 1/20$, $4/5$ at $\varepsilon = 1/10$, $3/5$ at $\varepsilon = 1/5$. Every certificate is exact (iff), the independent checker and the rational LP primal/dual strong-duality check both pass on the full 71-action panel.

5.3 T2c certified bound and near-tightness

For a separating position under $\varepsilon = 1/10$, the certified Hellinger information interval is $[0.510826, 0.510826]$. The certified sufficiency bound for $\mathbb{P}(\text{correct}) \geq 0.99$ requires $n = 8$ repeats, giving certified $\mathbb{P}(\text{correct}) \geq 0.9916$ and $\mathbb{P}(\text{wrong}) \leq 0.0084$. The exact minimax oracle (exhaustive two-class enumeration) reaches 0.99 at $n = 5$, so the certified bound is conservative by a small additive constant. The full sweep is in Table 1.

Table 1: add riboswitch: certified vs. exact sample complexity.

ε	certified I (per separating probe)	certified n for $\kappa = .99$	exact n (oracle)
1/20	0.830366	5	3
1/10	0.510826	8	5
1/5	0.223144	18	13

5.4 T2d costed design

With unit cost per read and the single pair $v = \theta_1 - \theta_0$, the costed design concentrates repeats on one separating position (the separating positions are information-equivalent). At $\varepsilon = 1/10$ the achievable integer cost is 8; the LP dual lower bound is below 4; the integrality gap is 4.5%. At budget 1 the design is a certified no-go (status `NO_GO`).

5.5 Executable

The certificates are emitted by `scripts/real_add_run.py` over the registered model in `src/d2t_rna/t2/real_add.py`; `tests/t2/test_real_add.py` locks the model construction, the noise channel, the finiteness/positivity of the Hellinger information, and the sufficiency bound. The certificate JSON is written to the artifacts run directory (path env-configurable via `D2T_RNA_ARTIFACTS_ROOT`). A self-contained reproducibility image is provided by `Dockerfile` (`docker build -t d2t-rna/paper-artifact .`); its build step installs the package from a clean copy, emits the real-data certificate, and runs the locked T2 suite, so the artifact is reproducible without any server-local state. The decision-theoretic certificate path is pure-Python stdlib; `pyproject.toml` additionally declares `numpy/scipy/ pydantic` for other package modules.

5.6 Measured observation channel: 1M7 SHAPE reactivity

The model-conditional case above uses a synthetic bit-flip coupling. We now register an observation law *derived from* measured reactivity from a published assay, used as a *registered finite observation model* (`DESCRIPTIVE_TRANSFORM / MODEL_SENSITIVITY_ONLY`). The reactivity only fixes the registered observation law; it is *not* treated as measured biological truth, and the certificates below are descriptive, model-conditional computations (`DESCRIPTIVE_ONLY`), not a real gamma certificate on biological data. Dataset (registered, CC0, RMDB accession `ADD71_STD_0001`): Tian S, Kladwang W, Das R, “Allosteric mechanism of the *V. vulnificus* adenine riboswitch resolved by four-dimensional chemical mapping,” *eLife* (2018) 7:e29602, doi:10.7554/eLife.29602, PMID 29446752. Construct: add riboswitch residues 13–83 (71 nt). Reagent: 1M7 (SHAPE), normalized so loop residues have mean reactivity 1.0. Two conformations are used: apo (`RDAT REACTIVITY:1`) and 5 mM adenine-bound (`RDAT REACTIVITY:2`).

Binarization. For each probed position u and conformation s , the readout is binary {paired/protected, unpaired/reactive}. The probability of reading *unpaired/reactive* is the measured normalized reactivity $r[s, u]$, clamped to a registered measurement floor 1/100,

$$p_{\text{reactive}}(s, u) = \max(1/100, \min(99/100, r[s, u])),$$

$$q(s, u) = 1 - p_{\text{reactive}}(s, u).$$

The 1% floor encodes the repeats/resolution of the measurement (the reported per-position standard errors show the class call is not sub-1% precise) and keeps the Hellinger information per separating position finite, so the finite-sample bound is genuine rather than a degenerate $n = 1$ certainty. Position u is *measured-separating* iff $q(\text{apo}, u) \neq q(\text{bound}, u)$; the separation coefficient for a single probe is $\gamma = |q(\text{bound}, u) - q(\text{apo}, u)|$.

Results. Of the 71 positions, 64 are measured-separating under the registered law. The registration mirrors contract 8.5 (role `REGISTERED_OBSERVATION_MODEL`): states = two conformations $\theta_0 = e_{\text{OFF}} = (1, 0)$, $\theta_1 = e_{\text{ON}} = (0, 1)$; passive marginal $M = (1, 1)$ so $M\theta_0 = M\theta_1 = 1$ (collision); one registered observation-model action per position. The T2b certificate over a panel of the eight strongest separators is exact (*iff*) with $\gamma = 49/50$; the rational LP and the enumeration checker agree and strong duality passes. For T2c, the strongest separator (position 10) certifies $\mathbb{P}(\text{correct}) \geq 0.99$ with $n = 3$ reads *under the registered observation model*; all 64 separating positions carry a certified finite sufficiency bound. For T2d, the achievable integer design repeats position 10 three times (certified cost 3); at budget $B = 1$ the design is a certified `NO_GO`, at $B = 3$ feasible. These n and cost figures are *model-conditional sample counts* (`DESCRIPTIVE_ONLY`) and are *not* claims about biological repeat counts, absolute probability, or wet-lab cost saving. The full per-position reactivity, error, binarized probabilities, and per-probe sufficiency bounds are emitted by `scripts/real_add_measured_run.py` over `src/d2t_rna/data/measured_add.py` (RDAT parsed from `src/d2t_rna/data/raw/ADD71_STD_0001.rdat`); `tests/data/test_measured_add.py` locks the RDAT parse, the stochastic channel, measured separation, and the finite-sample bound. The certificate is *model-conditional* throughout: the latent OFF/ON hypotheses and the binarization rule are registered, and the observation law is derived from measured in-vitro reactivity. The result is therefore a descriptive, model-conditional computation on a registered finite observation model (`DESCRIPTIVE_ONLY` / `MODEL_SENSITIVITY_ONLY`) — *not* a real gamma certificate on biological data, and the reactivity is not treated as measured biological truth.

Second real case: gcvT glycine riboswitch (DMS). To establish transferability we execute the identical workflow on a second, chemically different published assay. Dataset (registered, RMDB accessions `BSUGLY_DMS_0013` / `BSUGLY_DMS_0014`): the *Bacillus subtilis* gcvT glycine riboswitch $\Delta P0$ variant, co-transcriptionally folded, TECprobe-VL, DMS reactivity (DMS methylates solvent-exposed A/C), analyzed with ShapeMapper2 (Busan & Weeks, *RNA* 2017, doi:10.1261/rna.061945.117) and normalized per Low & Weeks (*Methods* 2010, doi:10.1016/j.ymeth.2010.06.007). The two conditions are 0 mM glycine (apo, `BSUGLY_DMS_0013`) and 1 mM glycine (bound, `BSUGLY_DMS_0014`); the full-length 265-nt transcript is used. The binarization rule $p_{\text{reactive}} = \max(1/100, \min(99/100, r))$ and the registration slots are identical to the add case; only the chemistry, ligand, and construct differ.

Results (second case). Of the 265 positions, 206 are measured-separating under the registered law. The T2b certificate over the eight strongest separators is exact (*iff*) with $\gamma > 0$ (rational

LP and enumeration checker agree, strong duality passes). For T2c, the strongest separator (position 221) certifies $\mathbb{P}(\text{correct}) \geq 0.99$ with $n = 15$ reads *under the registered observation model*, and all 206 separating positions carry a certified finite sufficiency bound. For T2d, the achievable integer design is again a certified NO_GO at budget $B = 1$ and feasible at larger B . Two chemistries, two ligands, and two constructs therefore register the identical certificate workflow. This demonstrates that the registration slots are reusable across these published assays; it is a DESCRIPTIVE_ONLY / MODEL_SENSITIVITY_ONLY observation, *not* a claim of cross-system transfer or external validation, and does not assert that the add-riboswitch result is free of assay-specific artifacts. Emitted by `scripts/real_glycine_measured_run.py` over `src/d2t_rna/data/measured_glycine.py` (RDAT parsed from `src/d2t_rna/data/raw/BSUGLY_DMS_0013.rdat` and `src/d2t_rna/data/raw/BSUGLY_DMS_0014.rdat`); `tests/data/test_measured_glycine.py` locks the RDAT parse, the stochastic channel, measured separation, the finite-sample bound, and the transferability difference from the add case.

Third real case: miniTTR designed metal-ion switch (DMS). To probe the remaining *riboswitch*- and *ligand-specificity* concern, we register a third measured case that is neither a natural aptamer nor ligand-gated. Dataset (registered, RMDB accession MTTR1_MGTI_0001): Yesselman JD et al., “Computational design of three-dimensional RNA structure and function,” *Nature Nanotechnology* (2019), doi:10.1038/s41565-019-0517-8, PMID 31427748. Construct: miniTTR c1, a small *designed* RNA (81 nt). Reagent: DMS reactivity measured over a 32-point MgCl_2 titration. The two conditions used are the depleted and saturated endpoints: 0 mM MgCl_2 (RDAT ANNOTATION_DATA:1) and 50 mM MgCl_2 (RDAT ANNOTATION_DATA:32). The binarization rule $p_{\text{reactive}} = \max(1/100, \min(99/100, r))$ and the registration slots are identical to the add/glycine cases; the discriminator is a *metal-ion*-driven conformational switch, not a ligand-occupancy switch.

Results (third case). Of the 81 positions, 76 are measured-separating under the registered law. The T2b certificate over the eight strongest separators is exact (iff) with $\gamma = 49/50$ (rational LP and enumeration checker agree, strong duality passes). For T2c, the strongest separator (position 23) certifies $\mathbb{P}(\text{correct}) \geq 0.99$ with $n = 3$ reads *under the registered observation model*, and all 76 separating positions carry a certified finite sufficiency bound. For T2d, the achievable integer design reads a certified cost of 3, and is a certified NO_GO at budget $B = 1$. Three published assays — a natural SHAPE riboswitch, a natural DMS riboswitch, and a *designed* DMS metal-ion switch — therefore all register the identical certificate workflow, which shows the registration slots apply across these inputs. This is a DESCRIPTIVE_ONLY / MODEL_SENSITIVITY_ONLY observation on registered finite observation models; it is *not* a claim of cross-system transfer, biological or design-provenance generalization, or a real gamma certificate on biological data. Emitted by `scripts/real_mattr_measured_run.py` over `src/d2t_rna/data/measured_mattr.py` (RDAT parsed from `src/d2t_rna/data/raw/MTTR1_MGTI_0001.rdat`); `tests/data/test_measured_mattr.py` locks the RDAT parse, the two titration endpoints, the stochastic channel, measured separation, the finite-sample bound, and the transferability differences from both prior cases.

5.7 Fail-closed negative control on measured data

To show the executable is *honest* (fail-closed, contract 8.4 / 10.2), we run a negative control on the same registered add observation model (RMDB ADD71_STD_0001, Section 5.6). The action panel is restricted to the seven positions whose binarized apo/bound reactivity read identical after the registered 1% readout floor, so each carries zero measured Hellinger information between the two classes. The *same* code path that emitted the add and glycine separation certificates now

certifies *collision* (Definition ??): $\gamma = 0$ with a non-trivial collision witness and no separation witness, cross-checked by exact enumeration and LP strong duality. The verdict is **FAIL_CLOSED**. This is not a synthetic counterexample: the observation law is derived from the measured 1M7 SHAPE reactivities of the published assay, and only the panel differs. It demonstrates that, on this registered model, the code path is fail-closed — it refuses to certify separation when the registered observation law does not support it. Emitted by `scripts/real_negative_control_run.py` over `src/d2t_rna/data/measured_negative.py`; `tests/data/test_measured_negative.py` locks the collision verdict, the shared-panel zero-information property, the LP/enumeration cross-check, and the non-contamination of the positive separation certificate.

6 Complete baseline table (88 rows)

7 Phase4-v2 synthetic benchmark tables

This section records the executed Phase4-v2 synthetic benchmark that supports the comparable-only headline and the sealed-family comparative in the main text (Section ??). All values are model-conditional synthetic results bound to the immutable artifacts under `/mnt/cunyuliu/d2t-rna/artifacts/phase4v2/` and the manifest `manifests/audit/v7_p4_comparative_v1.json`; no real-data, population, transfer, or biological claim is made.

7.1 80-cell catalog-pair grid

Table 3 lists every cell of the 80-cell grid (4 catalog classes **CA–CD** $\times$ 5 pairs $\times$ 2 budgets $\times$ 2 cost modes). For each cell it reports the exhaustive-oracle minimax error (the global minimum), the best same-task comparable error, and whether any baseline beat the oracle (none does: **oracle_never_beaten** on all 80 cells).

7.2 Non-equivalent-action ablation (12 cells)

Table 4 lists the 12 ablation cells. The identical-channel negative control is labeled **PRICE_SUBSTITUTION_OR_TIE_CONTROL** and is not read as action informativeness; on every cell the exact oracle is never beaten.

7.3 Sealed-family comparative (24 blocks)

Table 5 lists every sealed block. **D2T** is the exact decision framework (oracle-optimal); **Comparator** is the matched strongest comparable (Chernoff). d_i is per-block $L(\text{D2T}) - L(\text{Comparator})$; $d_i \leq 0$ everywhere (**d2t_never_worse_than_comparator**= true). Pooled per-block delta mean $\bar{d} = -0.000229$ with 10^4 -block bootstrap CI $[-0.000572, 0]$ (not strictly below zero). Exact certificates on 19/24 blocks; 2 strict improvements; 4 comparator blocks Pareto-dominated; 5 bound-only (fail-closed); 0 timeouts, 0 failures. This is a **COMPARATIVE_SYNTHETIC_RECORD**, not a population or real-data generalization claim.

Table 2: Complete 88-row model-conditional baseline summary (11 micro-cases $\times$ 8 baselines). Long rationals are truncated to 9 leading digits; full values are in the packaged TSV.

case	baseline	cost	product	TV	minmax	err	abstain	correct
alternating_rectangle_2x2	exhaustive_oracle	0	0	1/2	0	1/2	0	1/2
alternating_rectangle_2x2	full_matrix	8	0	1/2	0	1/2	0	1/2
alternating_rectangle_2x2	random	8	0	1/2	0	1/2	0	1/2
alternating_rectangle_2x2	greedy_test_cover	8	0	1/2	0	1/2	0	1/2
alternating_rectangle_2x2	eig	8	0	1/2	0	1/2	0	1/2
alternating_rectangle_2x2	chernoff	8	0	1/2	0	1/2	0	1/2
alternating_rectangle_2x2	lm2r_heuristic	8	0	1/2	0	1/2	0	1/2
alternating_rectangle_2x2	t2_integer_lp	8	0	1/2	0	1/2	0	1/2
boundary	exhaustive_oracle	6	63/64	1/128	0	127/128	0	127/128
boundary	full_matrix	8	255/256	1/512	0	511/512	0	511/512
boundary	random	8	255/256	1/512	0	511/512	0	511/512
boundary	greedy_test_cover	8	255/256	1/512	0	511/512	0	511/512
boundary	eig	8	255/256	1/512	0	511/512	0	511/512
boundary	chernoff	8	255/256	1/512	0	511/512	0	511/512
boundary	lm2r_heuristic	8	255/256	1/512	0	511/512	0	511/512
boundary	t2_integer_lp	1	1/2	1/4	0	3/4	0	3/4
cancellation_cycled	exhaustive_oracle	0	0	1/2	0	1/2	0	1/2
cancellation_cycled	full_matrix	8	0	1/2	0	1/2	0	1/2
cancellation_cycled	random	8	0	1/2	0	1/2	0	1/2
cancellation_cycled	greedy_test_cover	8	0	1/2	0	1/2	0	1/2
cancellation_cycled	eig	8	0	1/2	0	1/2	0	1/2
cancellation_cycled	chernoff	8	0	1/2	0	1/2	0	1/2
cancellation_cycled	lm2r_heuristic	8	0	1/2	0	1/2	0	1/2
cancellation_cycled	t2_integer_lp	8	0	1/2	0	1/2	0	1/2
exact_collision	exhaustive_oracle	0	0	1/2	0	1/2	0	1/2
exact_collision	full_matrix	8	0	1/2	0	1/2	0	1/2
exact_collision	random	8	0	1/2	0	1/2	0	1/2
exact_collision	greedy_test_cover	8	0	1/2	0	1/2	0	1/2
exact_collision	eig	8	0	1/2	0	1/2	0	1/2
exact_collision	chernoff	8	0	1/2	0	1/2	0	1/2
exact_collision	lm2r_heuristic	8	0	1/2	0	1/2	0	1/2
exact_collision	t2_integer_lp	8	0	1/2	0	1/2	0	1/2
near_collision	exhaustive_oracle	6	187686999..	998123130..	0	100187686..	0	100187686..
near_collision	full_matrix	8	437935747..	199562064..	0	200437935..	0	200437935..
near_collision	random	8	437935747..	199562064..	0	200437935..	0	200437935..
near_collision	greedy_test_cover	8	437935747..	199562064..	0	200437935..	0	200437935..
near_collision	eig	8	437935747..	199562064..	0	200437935..	0	200437935..
near_collision	chernoff	8	437935747..	199562064..	0	200437935..	0	200437935..
near_collision	lm2r_heuristic	8	437935747..	199562064..	0	200437935..	0	200437935..
near_collision	t2_integer_lp	8	437935747..	199562064..	0	200437935..	0	200437935..
no_cycle	exhaustive_oracle	1	1	0	0	1	0	1
no_cycle	full_matrix	8	1	0	0	1	0	1
no_cycle	random	8	1	0	0	1	0	1
no_cycle	greedy_test_cover	8	1	0	0	1	0	1
no_cycle	eig	8	1	0	0	1	0	1
no_cycle	chernoff	8	1	0	0	1	0	1
no_cycle	lm2r_heuristic	8	1	0	0	1	0	1
no_cycle	t2_integer_lp	1	1	0	0	1	0	1
nondecomposable_3d	exhaustive_oracle	0	0	1/2	0	1/2	0	1/2
nondecomposable_3d	full_matrix	8	0	1/2	0	1/2	0	1/2
nondecomposable_3d	random	8	0	1/2	0	1/2	0	1/2
nondecomposable_3d	greedy_test_cover	8	0	1/2	0	1/2	0	1/2
nondecomposable_3d	eig	8	0	1/2	0	1/2	0	1/2
nondecomposable_3d	chernoff	8	0	1/2	0	1/2	0	1/2
nondecomposable_3d	lm2r_heuristic	8	0	1/2	0	1/2	0	1/2
nondecomposable_3d	t2_integer_lp	8	0	1/2	0	1/2	0	1/2

Table 3: 80-cell catalog-pair grid. **Oracle** is the exhaustive minimax error (correctness reference, never ranked); **Best comparable** is the lowest minimax error among the executed same-task comparable labels; **Beaten** = any baseline strictly below the oracle.

class	pair	budget	cost	oracle	best comp	beaten
CA	CA_p1	4	hetero	81/512	27/128	no
CA	CA_p1	4	uniform	81/512	81/512	no
CA	CA_p1	8	hetero	6561/131072	243/2048	no
CA	CA_p1	8	uniform	6561/131072	6561/131072	no
CA	CA_p2	4	hetero	1/512	1/128	no
CA	CA_p2	4	uniform	1/512	1/512	no
CA	CA_p2	8	hetero	1/131072	1/131072	no
CA	CA_p2	8	uniform	1/131072	1/131072	no
CA	CA_p3	4	hetero	147/512	11/32	no
CA	CA_p3	4	uniform	147/512	147/512	no
CA	CA_p3	8	hetero	30539/131072	30539/131072	no
CA	CA_p3	8	uniform	30539/131072	30539/131072	no
CA	CA_p4	4	hetero	147/512	11/32	no
CA	CA_p4	4	uniform	147/512	147/512	no
CA	CA_p4	8	hetero	30539/131072	30539/131072	no
CA	CA_p4	8	uniform	30539/131072	30539/131072	no
CA	CA_p5	4	hetero	81/512	27/128	no
CA	CA_p5	4	uniform	81/512	81/512	no
CA	CA_p5	8	hetero	6561/131072	243/2048	no
CA	CA_p5	8	uniform	6561/131072	6561/131072	no
CB	CB_p1	4	hetero	81/512	27/128	no
CB	CB_p1	4	uniform	81/512	81/512	no
CB	CB_p1	8	hetero	6561/131072	243/2048	no
CB	CB_p1	8	uniform	6561/131072	6561/131072	no
CB	CB_p2	4	hetero	147/512	11/32	no
CB	CB_p2	4	uniform	147/512	147/512	no
CB	CB_p2	8	hetero	30539/131072	30539/131072	no
CB	CB_p2	8	uniform	30539/131072	30539/131072	no
CB	CB_p3	4	hetero	1/512	1/128	no
CB	CB_p3	4	uniform	1/512	1/512	no
CB	CB_p3	8	hetero	1/131072	1/131072	no
CB	CB_p3	8	uniform	1/131072	1/131072	no
CB	CB_p4	4	hetero	1/512	1/128	no
CB	CB_p4	4	uniform	1/512	1/512	no
CB	CB_p4	8	hetero	1/131072	1/131072	no
CB	CB_p4	8	uniform	1/131072	1/131072	no
CB	CB_p5	4	hetero	147/512	11/32	no
CB	CB_p5	4	uniform	147/512	147/512	no
CB	CB_p5	8	hetero	30539/131072	30539/131072	no
CB	CB_p5	8	uniform	30539/131072	30539/131072	no
CC	CC_p1	4	hetero	1/32	1/16	no
CC	CC_p1	4	uniform	1/32	1/32	no
CC	CC_p1	8	hetero	1/512	1/512	no
CC	CC_p1	8	uniform	1/512	1/512	no
CC	CC_p2	4	hetero	1/32	1/2	no
CC	CC_p2	4	uniform	1/32	1/32	no
CC	CC_p2	8	hetero	1/512	1/16	no
CC	CC_p2	8	uniform	1/512	1/32	no
CC	CC_p3	4	hetero	0	0	no
CC	CC_p3	4	uniform	0	0	no
CC	CC_p3	8	hetero	0	0	no
CC	CC_p3	8	uniform	0	0	no
CC	CC_p4	4	hetero	0	0	no
CC	CC_p4	4	uniform	0	0	no
CC	CC_p4	8	hetero	0	0	no
CC	CC_p4	8	uniform	9 0	0	no
CC	CC_p5	4	hetero	1/32	1/16	no
CC	CC_p5	4	uniform	1/32	1/32	no
CC	CC_p5	8	hetero	1/512	1/4	no
CC	CC_p5	8	uniform	1/512	1/4	no

Table 4: Non-equivalent-action ablation. **Interpretation** is the pre-registered interpretation lock per panel.

cell	panel	budget	cost mode	interpretation	oracle err	beaten
A_identical_control_b4_x_hetero	A_identical_control	4	hetero	PRICE_SUBSTITUTION_OR_TIE_CONTROL	5/32	no
A_identical_control_b4_x_uniform	A_identical_control	4	uniform	PRICE_SUBSTITUTION_OR_TIE_CONTROL	5/32	no
A_identical_control_b8_x_hetero	A_identical_control	8	hetero	PRICE_SUBSTITUTION_OR_TIE_CONTROL	289/4096	no
A_identical_control_b8_x_uniform	A_identical_control	8	uniform	PRICE_SUBSTITUTION_OR_TIE_CONTROL	289/4096	no
B_crossing_informativeness_b4_x_hetero	B_crossing_informativeness	4	hetero	NON_EQUIVALENT_ACTION_RISK_COST_TRADEOFF	1/10	no
B_crossing_informativeness_b4_x_uniform	B_crossing_informativeness	4	uniform	NON_EQUIVALENT_ACTION_RISK_COST_TRADEOFF	7/250	no
B_crossing_informativeness_b8_x_hetero	B_crossing_informativeness	8	hetero	NON_EQUIVALENT_ACTION_RISK_COST_TRADEOFF	8/125	no
B_crossing_informativeness_b8_x_uniform	B_crossing_informativeness	8	uniform	NON_EQUIVALENT_ACTION_RISK_COST_TRADEOFF	341/125000	no
C_3state_identity_pair_b4_x_hetero	C_3state_identity_pair	4	hetero	NON_EQUIVALENT_ACTION_CHANNEL_RANK	1/32	no
C_3state_identity_pair_b4_x_uniform	C_3state_identity_pair	4	uniform	NON_EQUIVALENT_ACTION_CHANNEL_RANK	1/32	no
C_3state_identity_pair_b8_x_hetero	C_3state_identity_pair	8	hetero	NON_EQUIVALENT_ACTION_CHANNEL_RANK	1/512	no
C_3state_identity_pair_b8_x_uniform	C_3state_identity_pair	8	uniform	NON_EQUIVALENT_ACTION_CHANNEL_RANK	1/512	no

Table 5: Sealed-family comparative. **Exact** is the per-block exact status (RUN = exact certificate, BOUND = bound-only); **D2T** / **Cmp** are the D2T and comparator minimax errors; d_i is the per-block delta; **Strict** = strict improvement; **Dom** = comparator Pareto-dominated.

family	block	exact	D2T err	Cmp err	d_i	Strict	Dom
q_s2_2x2_noisy	q_s2_2x2_b1	RUN	214517/625000	214517/625000	0	no	no
q_s2_2x2_noisy	q_s2_2x2_b2	RUN	183229/500000	183229/500000	0	no	yes
q_s2_2x2_noisy	q_s2_2x2_b3	RUN	3356401/12500000	3356401/12500000	0	no	no
q_s2_2x2_noisy	q_s2_2x2_b4	RUN	214517/625000	214517/625000	0	no	no
q_s2_3x3_noisy	q_s2_3x3_b1	RUN	143797874333033/625000000000000	143797874333033/625000000000000	0	no	no
q_s2_3x3_noisy	q_s2_3x3_b2	RUN	17727014423477/625000000000000	17727014423477/625000000000000	0	no	no
q_s2_3x3_noisy	q_s2_3x3_b3	RUN	1546903518469/500000000000000	1546903518469/500000000000000	0	no	yes
q_s2_3x3_noisy	q_s2_3x3_b4	RUN	17727014423477/625000000000000	17727014423477/625000000000000	0	no	no
q_s2_3x3_noisy	q_s2_3x3_b5	RUN	143797874333033/625000000000000	143797874333033/625000000000000	0	no	no
q_s3_2x2_noisy	q_s3_2x2_b1	RUN	9558413/37500000	9558413/37500000	0	no	no
q_s3_2x2_noisy	q_s3_2x2_b2	RUN	12085219/37500000	12085219/37500000	0	no	no
q_s3_2x2_noisy	q_s3_2x2_b3	RUN	9558413/37500000	9558413/37500000	0	no	no
q_s3_2x2_noisy	q_s3_2x2_b4	RUN	12085219/37500000	12085219/37500000	0	no	no
q_s3_2x2_noisy	q_s3_2x2_b5	RUN	12443887/37500000	12443887/37500000	0	no	no
q_s3_3x3_noisy	q_s3_3x3_b1	RUN	9558413/37500000	9558413/37500000	0	no	no
q_s3_3x3_noisy	q_s3_3x3_b2	RUN	4321315661/135000000000	12085219/37500000	-29363179/13500000000	yes	yes
q_s3_3x3_noisy	q_s3_3x3_b3	RUN	9558413/37500000	9558413/37500000	0	no	no
q_s3_3x3_noisy	q_s3_3x3_b4	RUN	4321315661/135000000000	12085219/37500000	-29363179/13500000000	yes	yes
q_s3_3x3_noisy	q_s3_3x3_b5	RUN	12443887/37500000	12443887/37500000	0	no	no
q_s3_3x2_noisy_unsolv	q_s3_3x2_b1	BOUND_ONLY	—	9558413/37500000	—	no	no
q_s3_3x2_noisy_unsolv	q_s3_3x2_b2	BOUND_ONLY	—	12085219/37500000	—	no	no
q_s3_3x2_noisy_unsolv	q_s3_3x2_b3	BOUND_ONLY	—	9558413/37500000	—	no	no
q_s3_3x2_noisy_unsolv	q_s3_3x2_b4	BOUND_ONLY	—	12085219/37500000	—	no	no
q_s3_3x2_noisy_unsolv	q_s3_3x2_b5	BOUND_ONLY	—	12443887/37500000	—	no	no